Latent Roses: Capturing Topological Features with Autoencoder Activations

Joan Sebastian Salazar Montoya and Manuela Bastidas Olivares

Abstract An autoencoder is a special kind of neural network architecture that is trained such that the output is as close as possible to its input. Autoencoders are designed using a bottleneck architecture, which forces them to extract useful features from the data. To analyze if the learned features capture the topology of the data, we trained a convolutional autoencoder on a synthetic dataset of images with known topology. Then we collect the activations of the neurons in the bottleneck layer, interpret them as points in a 64D space and do a UMAP 2D projection of this point-cloud. We find that the UMAP projection captures the topology of the synthetic dataset.

1 Introduction

A neural network is a parametric map $f_\theta : \mathbb{R}^n \to \mathbb{R}^m$ defined by the composition of parametric functions $f^1_{\theta_1}, \ldots, f^L_{\theta_L}$, such that $f_\theta = f^L_{\theta_L} \circ \cdots \circ f^1_{\theta_1}$, where $\theta = (\theta_1, \ldots, \theta_L)$ denotes the full collection of parameters [6]. Neural networks are used to approximate a function $f^* : \mathbb{R}^n \to \mathbb{R}^m$ for which we only have noisy input-output pairs $(x, f^*(x))$ given in the form of a dataset, which is divided into a training set and a test set, where the latter is used for measuring the generalization capabilities of the neural network. Training the neural network consists of minimizing a loss function $\mathcal{L}(\theta)$, which is a measure of how much the neural network f_θ deviates from the desired function f^* over the training set. The training process is performed by

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using some variant of gradient descent [4]. Once trained, the neural network f_θ can be evaluated on new previously unseen samples, a process known as inference.

The choice for $f^1_{\theta_1}, \ldots, f^L_{\theta_L}$ is called an architecture of the neural network. A specific architecture for neural networks is called an autoencoder [6]. In this variant, the function is a map with the same domain and codomain $f : \mathbb{R}^n \to \mathbb{R}^n$ and the function one wants to approximate is the identity function. The loss function in this case is usually given by Mean Square Error (MSE) over the outputs of the neural network:

$$\mathcal{L}(\theta) = \sum_{x \in \mathcal{D}} \|x - f(x)\|^2. \tag{1}$$

Autoencoders are usually designed so that they consist of the composition of an encoder E and a decoder network D, so that for a sample x, $f(x) = D(E(x))$ [6]. The encoder E is usually designed to perform a dimensionality reduction of the input, so that $E : \mathbb{R}^n \to \mathbb{R}^k$ is a mapping where $k < n$. The decoder D is then a map $D : \mathbb{R}^k \to \mathbb{R}^n$. This encoder–decoder architecture forces the autoencoder to extract meaningful features from the dataset [3, 6, 2, 9]. When the dataset consists of images, it is useful to consider a convolutional autoencoder [1], where the first layers of the encoder are given by convolutional layers that perform convolutions over the images using learned kernels, and the last layers of the decoder do up-sampling by performing transposed convolutions. The objective of this work is to inspect the activations of the bottleneck layer of an autoencoder trained on synthetic images by performing dimensionality reduction over these activations. The pipeline of the process is depicted in Figure 1.

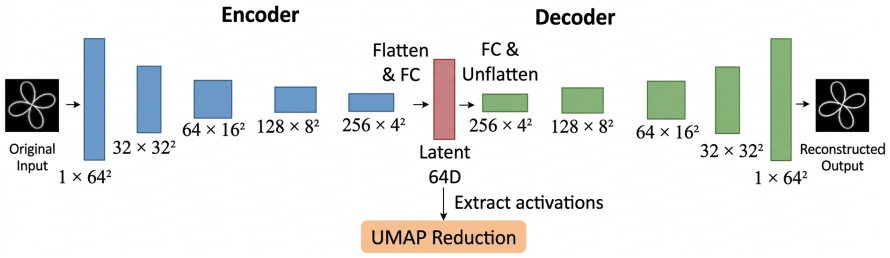


Fig. 1 Schematic of the experimental pipeline: rose-curve images are passed through the convolutional autoencoder; bottleneck activations are extracted and projected to 2D via UMAP.

2 Experiment

We created a synthetic dataset of 64×64 grayscale images of rose curves. These curves are given by the polar equation $r = \cos(k\theta + \phi)$. If k is odd the curve produces

k petals, and if k is even it produces $2k$ petals. The parameter ϕ controls the rotation of the curve. Some samples with $\phi = 0$ can be seen in Figure 2.

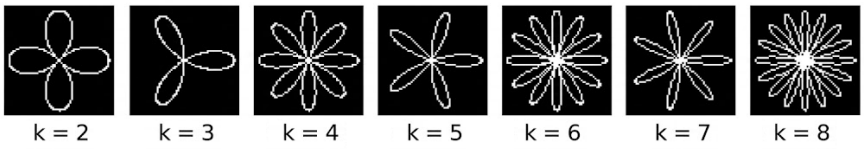


Fig. 2 Representative samples from the synthetic rose-curve dataset for each value of $k \in \{2, \ldots, 8\}$ at zero rotation. Each 64×64 image is a parametric curve given by the equation $r = \cos(k\theta)$.

We now describe the topological structure of the dataset. Note that ϕ is the only free continuous parameter after fixing k. Therefore, every rose-curve is labeled by a single real number. Moreover, since $r = \cos(k\theta + \phi)$ is a periodic function of ϕ, the space wraps around itself. Hence, for a fixed value of k, the set of curves in $\mathbb{R}^2$ satisfying the polar equation $r = \cos(k\theta + \phi)$ can be identified, via standard results in differential geometry (see, e.g., [7], [8]), with a smooth 1-dimensional manifold. One verifies that this manifold is both compact and connected; it then follows from the classification of compact connected 1-manifolds without boundary that it is diffeomorphic to S^1. Consequently, for m distinct choices of k, the resulting dataset is supported on a 1-dimensional manifold with m connected components, each of which is diffeomorphic to the circle. Note that the $1D$-manifold is *not* the curve displayed in each image (which is not even a manifold since it is not locally euclidean at the center), but the *set* of curves that satisfy the polar equation $r = \cos(k\theta + \phi)$.

We trained a symmetric convolutional autoencoder with four stride-2 convolutional blocks (channels $1 \to 32 \to 64 \to 128 \to 256$) followed by a fully connected bottleneck of dimension $d = 64$, and a mirrored decoder of four transposed convolutions with a sigmoid output, over the previous synthetic dataset. The parameter space used is $\{2, \ldots, 8\} \times [0, 2\pi]$ where the first set is the k value and the second set is the theta value. The loss for each sample is

$$\mathcal{L}(x, f(x)) = \frac{1}{N^2} \sum_{i=1}^{N^2} (x_i - f(x)_i)^2 . \tag{2}$$

that is, mean-squared error pixel-by-pixel. In this case, $N = 64$. Once trained, we performed inference over more samples from the dataset and extracted the corresponding activations of the bottleneck layer. Finally we performed UMAP dimensionality reduction [10] over this 64D point-cloud to 2D.

3 Results

Figure 3 shows the 2D UMAP projections of the bottleneck activations together with representative images from three of the seven clusters. blueThe UMAP parameters are $n_components = 2$, $n_neighbors = 30$, $min_dist = 0.1$, $random_state = 41$. The projection reveals seven distinct clusters, one for each value of $k \in \{2, \ldots, 8\}$. Each cluster groups images sharing the same number of petals, and the points within each cluster trace a roughly circular path corresponding to varying rotation angle θ.

The two-dimensional projections exhibit circular arrangements organized into discrete clusters, suggesting that the autoencoder is successfully capturing the underlying 1-manifold structure of the data. This is further confirmed by a k-NN classification accuracy of 100% on the projected representations ($k = 10$), demonstrating that the encoding preserves the inter-class separability of the original data. Moreover, the UMAP projection clearly reveals the expected S^1-like topology, with each cluster tracing a closed loop consistent with the expected structure.

We remark that a rigorous validation of this topological observation would require a formal quantification via persistent homology; however, the visual evidence provided by the UMAP projection is sufficient to support this geometric interpretation in the present context.

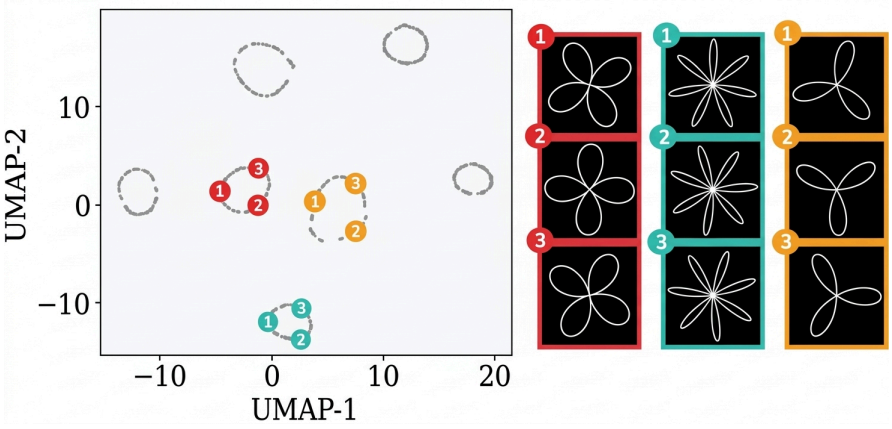


Fig. 3 (*Left*) 2D UMAP projections of the 64-dimensional bottleneck activations, with three sampled frame positions highlighted per cluster. (*Right*) Rose-curve images at the three sampled positions for three selected clusters, illustrating that each cluster groups curves sharing the same number of petals at varying rotations. The images are up-scaled for visualization.

4 Future Work

The circle-like appearance of the projections suggests that computing the persistent homology [5] of the activations should yield barcodes related to the inherent topology of the dataset. Motivated by this observation, our next step is to consider datasets without evident topological features (e.g. higher dimensional manifolds or non-orientable surfaces) where a formal TDA pipeline becomes essential.

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